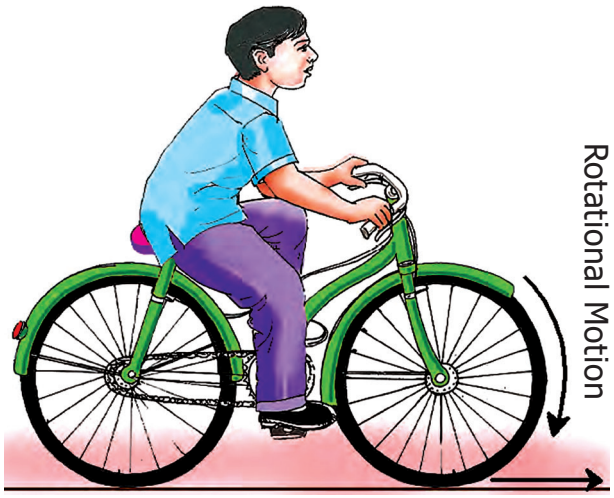
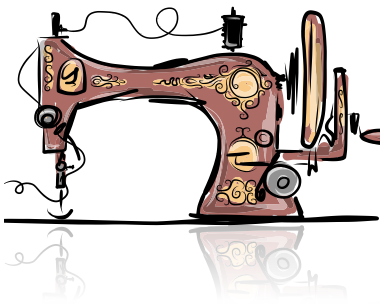


Linear Motion

Multiple motion in a sewing machine



- Motion of the needle

  _____

- Motion of the wheel

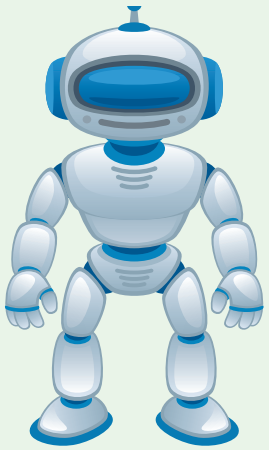
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- Motion of footrest

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2.3 Science Today - Robot

Robots are automatic machines. Some robots can perform mechanical and repetitive jobs faster and more accurately than people. Robots can also handle dangerous materials and explore distant planets.



A bipedal humanoid robot.

The term 'robot' comes from a czech word, 'robota' meaning 'forced labour'. Robotics is the science and study of robots.

What can Robots do?

Robots can sense and respond to their surroundings. They can handle delicate objects or apply great force. For example, they can perform eye operations guided by a human surgeon, or assemble a car. With **artificial intelligence**, robots will also be able to make decisions for themselves.

How do Robots sense?

Electronic sensors function as robot's eyes and ears. Twin video cameras give the robot a 3-D view of the world. Microphones detect sounds. Pressure



The quadrupedal military robot